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ECB1323-ANTENNA AND MILLIMETER WAVE COMMUNICATION

**DESIGN OF AN EFFICIENT MULTIBAND ANTENNA FOR
TERRESTRIAL APPLICATIONS**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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This Project Work ECB1323 – Antenna and Millimeter Wave Communication report has been submitted for the End Semester Project viva voce Examination held on

_____.

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ABSTRACT

The rapid growth of modern wireless communication systems has created a strong demand for compact, efficient, and multiband antennas capable of supporting multiple frequency standards within a single device. This work presents the design of an efficient multiband antenna tailored for terrestrial communication applications such as mobile networks, wireless broadband, and IoT systems. The proposed antenna is engineered to operate across multiple frequency bands while maintaining high radiation efficiency, stable gain, and compact size, making it suitable for integration into modern communication devices. The antenna design utilizes a microstrip patch configuration due to its low profile, ease of fabrication, and compatibility with planar structures. To achieve multiband functionality, techniques such as slot incorporation, stub loading, and geometric modification are employed. The substrate material and feeding technique are carefully selected to optimize impedance matching and bandwidth performance across all operating bands. Simulation and analysis are carried out using advanced electromagnetic design tools to evaluate key performance parameters including return loss, Voltage Standing Wave Ratio (VSWR), gain, radiation pattern, and bandwidth. The results demonstrate that the proposed antenna achieves satisfactory impedance matching ($VSWR < 2$) at all target frequencies, along with stable omnidirectional radiation characteristics, which are essential for terrestrial communication systems.

Abstract (Key words)	POs Mapping
The rapid growth of modern wireless communication systems has created a strong demand for compact, efficient, and multiband antennas capable of supporting multiple frequency standards within a single device.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO2

SDG Goal		Remarks
SDG 9 	Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.	Supports SDG 9 by enhancing technological development and improving efficient, innovative engineering solutions.

Project Component	Relevant IEEE Standards
Antenna design & wireless communication	IEEE 802.11, IEEE 802.16, IEEE 145
Multiband frequency operation & spectrum usage	IEEE 1900, IEEE 802.22
RF performance & antenna measurement	IEEE 149, IEEE 1128
IoT integration & smart communication systems	IEEE 2413, IEEE 802.15.4
Simulation & electromagnetic analysis	IEEE 1597

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
IoT	Internet of things
MIMO	Multi input and multi output
RF	Radio Frequency
GSM	Global System for Mobile Communications
LTE	Long Term Evolution
VSWR	Voltage Standing Wave Ratio

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION FOR 5G COMMUNICATION

Terrestrial applications refer to communication systems that operate on the Earth's surface, where both transmitting and receiving devices are located on land. These systems form the backbone of modern wireless communication and are widely used in applications such as mobile networks, wireless internet, broadcasting, and Internet of Things (IoT) devices. Unlike satellite communication, terrestrial systems rely on ground-based infrastructure such as base stations, routers, and antennas to transmit and receive signals over short to medium distances. In terrestrial communication, electromagnetic waves propagate through the atmosphere and are influenced by environmental factors such as buildings, terrain, weather conditions, and other obstacles. These factors can cause signal degradation due to reflection, diffraction, and scattering. Therefore, the design of antennas for terrestrial applications requires careful consideration of parameters like gain, bandwidth, efficiency, and radiation pattern to ensure reliable communication. With the rapid advancement of technologies such as 4G, 5G, and upcoming 6G systems, there is an increasing demand for compact and efficient multiband antennas capable of supporting multiple frequency bands simultaneously. Terrestrial applications require antennas that provide stable performance, wide coverage, and minimal interference across different communication standards. As a result, modern antenna design focuses on achieving high efficiency, compact size, and multiband operation to meet the growing needs of terrestrial wireless systems.

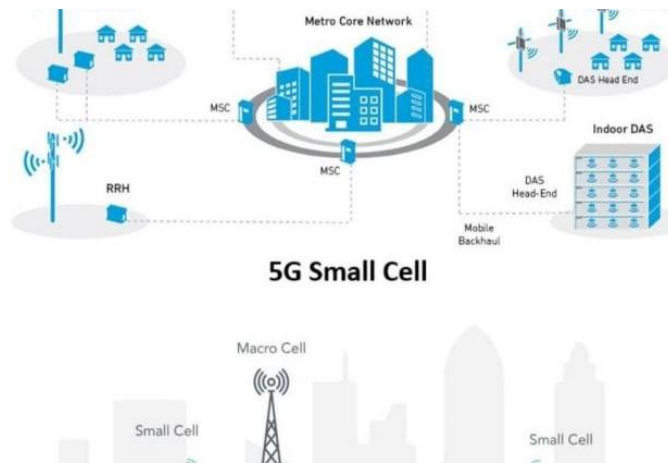


Figure 1.1 5G communication

1.2 5G NETWORKS

Terrestrial communication systems play a vital role in modern wireless technology by enabling communication over the Earth's surface through the use of antennas and RF devices. These systems are widely applied in cellular and mobile communication networks such as 4G and 5G, where base stations and user devices communicate efficiently across multiple frequency bands to provide high-speed voice and data services. In addition, wireless internet systems like Wi-Fi rely on terrestrial antennas to deliver connectivity in homes, offices, and public spaces, often operating across multiple bands to ensure better coverage and reduced interference. Furthermore, terrestrial communication is a key component in the development of smart cities and Internet of Things (IoT) applications, where sensors, smart devices, and automated systems communicate seamlessly to improve urban infrastructure and daily life. Broadcasting systems, including radio and television transmission, also depend on terrestrial antennas to provide wide-area coverage and reliable signal distribution.



Figure 1.2 Wi-Fi rely



Figure 1.2.1 terrestrial antennas

Furthermore, figure 1.2 terrestrial communication is a key component in the development of smart cities and Internet of Things (IoT) applications, where sensors, smart devices, and automated systems communicate seamlessly to improve urban infrastructure and daily life. Broadcasting systems, including radio and television transmission, also depend on terrestrial antennas to provide wide-area coverage and reliable signal distribution. In figure 1.2.1 defense and security applications, radar and surveillance systems use terrestrial communication to detect objects, monitor environments, and ensure national safety.

1.3 ELECTROMAGNETIC SPECTRUM

Multiband antennas are essential in a wide range of terrestrial communication applications due to their flexibility and efficiency. In mobile communication systems, they enable smartphones to support multiple technologies such as GSM, LTE, and 5G within a single device. In wireless networking, they are used in Wi-Fi routers and access points to provide stable internet connectivity across multiple frequency bands like 2.4 GHz and 5 GHz. In the field of Internet of Things (IoT), multiband antennas support communication between smart devices such as sensors, wearables, and home

automation systems. Additionally, broadcasting systems utilize these antennas for radio and television transmission, ensuring wide coverage and signal reliability. In military and defense applications, multiband antennas are used for secure communication and surveillance systems operating across different frequency ranges. Their ability to handle multiple bands simultaneously makes them highly suitable for modern communication needs.

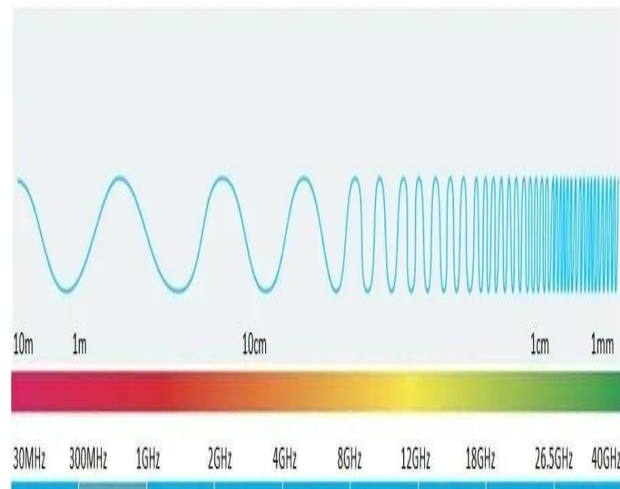


Figure 1.3 electromagnetic spectrum

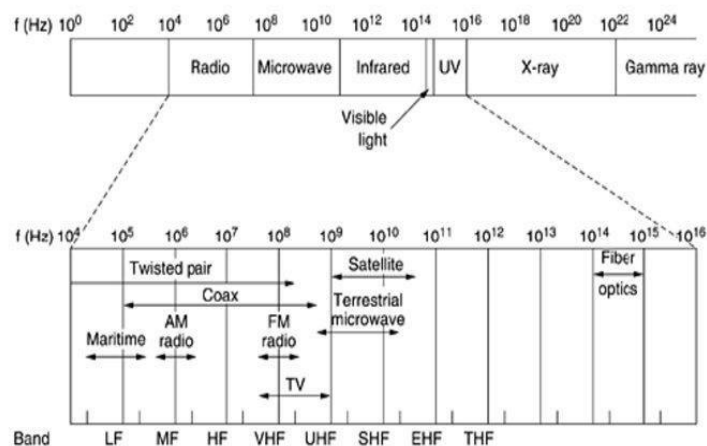


Figure 1.3.1 spectrum analysis

Multiband antennas figure 1.3 ,1.3.1 and operate across different regions of the electromagnetic spectrum, enabling communication over multiple frequencies.

1.4 RF AND RF DEVICES

RF devices are electronic components that operate within the radio frequency range and are essential for wireless communication systems. These devices are responsible for generating, transmitting, receiving, and processing RF signals. A typical RF system includes components such as transmitters, which convert baseband signals into RF signals for transmission; receivers, which capture RF signals and convert them back into usable electrical signals; and transceivers, which combine both functions in a single unit. Other important RF components include amplifiers, which increase signal strength, and filters, which remove unwanted frequencies and noise. In multiband antenna systems, RF devices must be designed to handle multiple frequency bands efficiently without causing interference.

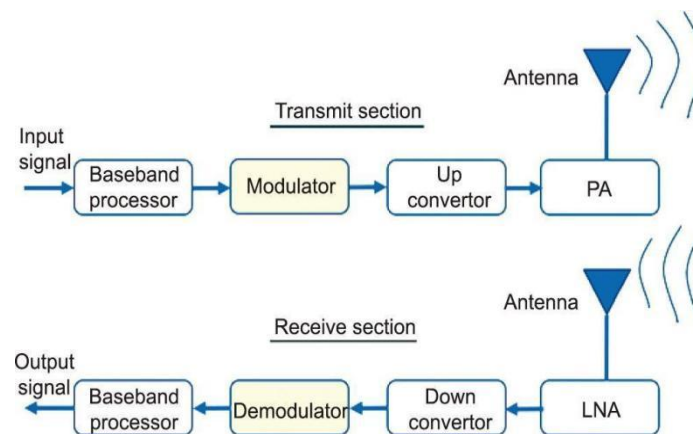


Figure 1.4 RF device

Proper integration of figure 1.4 RF devices with antennas ensures optimal system performance, improved signal quality, and reduced power consumption.

These devices are widely used in applications such as mobile phones, wireless routers, radar systems, and satellite communication.

1.5 ANTENNA

An antenna is a critical component in any wireless communication system, acting as a transducer that converts electrical signals into electromagnetic waves and vice versa. It enables the transmission of information through free space without the need for physical connections. There are various types of antennas, including dipole antennas, monopole antennas, loop antennas, and microstrip patch antennas. Among these, microstrip patch antennas are widely used in modern communication systems due to their compact size, low weight, ease of fabrication, and compatibility with printed circuit board (PCB) technology. For multiband operation, antennas are designed using techniques such as slotting, fractal geometry, and the addition of parasitic elements. These methods create multiple resonant frequencies within a single structure. The antenna design must also consider factors such as substrate material, feeding technique, and geometry to achieve the desired performance.

1.6 TYPES OF ANTENNA

1.6.1 Dipole Antenna

A dipole antenna is one of the simplest and most widely used antennas in wireless communication systems. It consists of two equal conductive elements (usually metal rods or wires) connected at the center by a feed line, where the RF signal is applied. The most common type is the half-wave dipole, whose total length is approximately half of the wavelength of the operating frequency. When current flows through the antenna, it creates an electromagnetic field that radiates energy into space, enabling signal transmission.

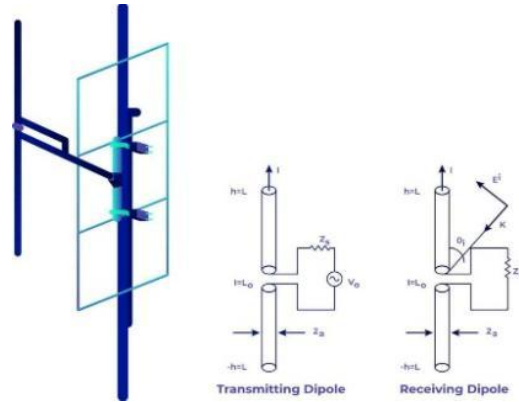


Figure 1.6.1 dipole antenna

1.6.2 Monopole Antenna

A monopole antenna is a type of antenna that consists of a single conducting rod or wire mounted vertically over a ground plane. It is essentially half of a dipole antenna, where the ground plane acts as a reflective surface that forms the missing half of the antenna. The most common form is the quarter-wave monopole antenna, whose length is approximately one-fourth of the operating wavelength. When RF current is applied at the feed point, the antenna radiates electromagnetic waves, and the ground plane helps in reflecting the waves to improve radiation efficiency.



Figure 1.6.2 Monopole Antenna

1.6.3 Microstrip Antenna

A microstrip patch antenna is a low-profile, lightweight, and compact antenna that is widely used in modern wireless communication systems. It consists of a metallic patch placed on one side of a dielectric substrate, with a ground plane on the other side. The patch is usually made of copper or gold and can have different shapes such as rectangular, circular, or triangular depending on the design requirements. The antenna operates by radiating electromagnetic waves when an RF signal is applied through a feeding technique such as microstrip line feed or coaxial probe feed. Microstrip patch antennas are popular due to their easy fabrication, low cost, and ability to be integrated with microwave integrated circuits .

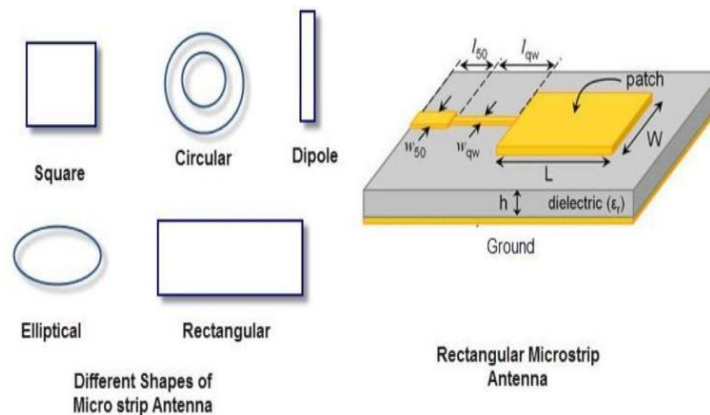


Figure 1.6.3 Microstrip Patch Antenna

1.6.4 Yagi-Uda Antenna

A Yagi–Uda antenna is a highly directional antenna widely used in terrestrial communication systems for long-distance signal reception and transmission. It consists of multiple parallel elements arranged on a single boom, including one driven element, one reflector, and several directors. The driven element is connected to the RF source and is responsible for radiating or

receiving signals, while the reflector is placed behind it to reflect signals forward, and the directors are placed in front to enhance and focus the radiation in a specific direction. This structure provides high gain and strong directivity compared to simple antennas like dipole or monopole antennas.

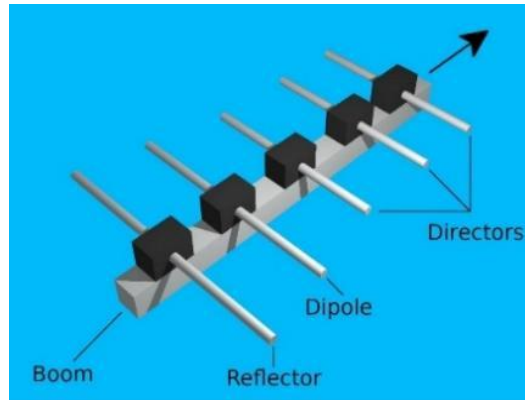


Figure 1.6.4 Yagi-Uda Antenna

A Figure 1.6.4 Yagi-Uda Antenna is a directional antenna widely used for radio and television signal reception. It consists of a driven element, a reflector, and one or more director elements arranged in a line. The antenna focuses energy in one direction, providing high gain and improved signal strength. It is commonly used in communication systems where long-distance and directional transmission is required.

1.6.5 Parabolic Reflector Antenna

A parabolic reflector antenna is a high-gain, highly directional antenna used for long-distance terrestrial and satellite communication systems. It consists of a curved parabolic-shaped reflector and a feed antenna (usually a horn antenna) placed at the focal point of the parabola. When RF signals are transmitted, the feed antenna radiates energy toward the reflector, which then reflects and concentrates the waves into a narrow, highly focused beam.



Figure 1.6.5 Parabolic Reflector Antenna

Similarly, during reception, Figure 1.6.5 Parabolic Reflector Antenna incoming parallel signals are reflected toward the feed point, improving signal strength and directivity. Due to its narrow beamwidth and high gain, this antenna is capable of transmitting and receiving signals over very long distances with minimal loss.

1.6.6 Array Antenna

An array antenna is a system in which multiple individual antenna elements are arranged in a specific geometric configuration such as linear, planar, or circular to achieve improved radiation performance. Each element in the array radiates electromagnetic waves, and when these waves combine, they produce a stronger and more directional radiation pattern compared to a single antenna. By properly controlling the amplitude and phase of the input signal to each element, the antenna beam can be steered in a desired direction, which is known as beamforming.



Figure 1.6.6 Array Antenna

The Figure 1.6.6 Array antennas provide high gain, better directivity, and improved signal quality, making them highly suitable for modern terrestrial communication systems.

1.7 ANTENNA PARAMETERS

1.7.1 Effective dielectric:

$$\epsilon_{eff} = \frac{\epsilon_R + 1}{2} + \frac{\epsilon_R - 1}{2} \left[\frac{1}{\sqrt{1 + 12 \left(\frac{h}{W} \right)}} \right]$$

The effective dielectric constant (ϵ_{eff}) is an important parameter in microstrip and patch antenna design that represents the combined effect of the substrate material and the surrounding air. Since electromagnetic waves in a microstrip antenna partially travel through the dielectric substrate and partially through air, the effective dielectric constant is always less than the actual dielectric constant of the substrate. It depends on the substrate height, patch width, and dielectric material used.

1.7.2 Length:

$$L = \frac{c}{2f_0\sqrt{\epsilon_{eff}}} - 0.824h \left(\frac{(\epsilon_{eff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\epsilon_{eff}-0.258)\left(\frac{W}{h}+0.8\right)} \right)$$

The length of an antenna is a very important parameter that decides its resonant frequency and overall performance. In most antennas, the physical length is related to the wavelength of the operating frequency. For example, in a half-wave dipole antenna, the length is approximately $\lambda/2$, while in a monopole antenna it is $\lambda/4$. In microstrip patch antennas, the length of the patch determines the resonance condition and is usually designed based on the effective dielectric constant. If the antenna length is not properly chosen, impedance mismatch occurs, leading to poor radiation efficiency and higher reflection losses. Therefore, accurate calculation of antenna length is essential for achieving proper resonance, good bandwidth.

1.7.3 Reflection Coefficient:

$$\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$$

Where:

Z_L = Load impedance (antenna)

Z_0 = Characteristic impedance (usually 50 Ω)

The reflection coefficient (S11) indicates how much of the input RF signal is reflected back from the antenna due to impedance mismatch. It is used to measure antenna performance and matching quality. A lower S11 value (below -10 dB) means better impedance matching, higher radiation efficiency, and less signal loss. In antenna design, minimizing the reflection coefficient is important for achieving good bandwidth and efficient operation in wireless communication systems.

1.7.4 Return Loss (S11) :

$$S_{11}=20\log\left(\frac{V_{reflected}}{V_{incident}}\right)$$

Loss indicates how much power is reflected back from the antenna due to impedance mismatch. A higher negative value (e.g., -10 dB or lower) indicates better matching and efficient power transfer.

1.7.5 VSWR:

$$VSWR = \frac{1+\Gamma}{1-\Gamma}$$

VSWR measures how efficiently RF power is transmitted from the transmission line to the antenna.

Ideal: VSWR = 1; Practical: VSWR $\leq$ 2

1.7.6 Gain:

$$G = \eta \times D$$

Where:

η = Efficiency

D = Directivity

Gain represents how well the antenna converts input power into radio waves in a specific direction. Higher gain means better signal strength and coverage.

1.7.7 Bandwidth:

$$BW = f_{high} - f_{low}$$

Where:

fH = Higher frequency

fL= Lower frequency

Bandwidth defines the range of frequencies over which the antenna

performs effectively (usually $VSWR \leq 2$ or $RL \leq -10$ dB).

1.7.8 Width

$$W = \frac{c}{2f_o \sqrt{\frac{\epsilon_R + 1}{2}}}$$

Width (W) in antenna design, especially in microstrip patch antennas, refers to the physical breadth of the radiating patch. It plays an important role in controlling the antenna's input impedance, radiation efficiency, and bandwidth. Proper selection of width helps improve power radiation and overall antenna performance, making the antenna more efficient for wireless communication applications.

CHAPTER 2

MICROSTRIP

2.1 INTRODUCTION TO MICROSTRIP ANTENNA

A microstrip antenna, also known as a patch antenna, is one of the most widely used antennas in modern wireless communication systems due to its compact size, lightweight structure, and ease of fabrication. It consists of a metallic patch placed on one side of a dielectric substrate, with a ground plane on the other side. The patch is typically made of conducting material such as copper or gold, while the substrate is made of dielectric materials like FR4, Rogers, or Duroid. The shape of the patch can vary, including rectangular, circular, triangular, or elliptical forms, depending on the application requirements. Among these, the rectangular microstrip patch antenna is the most commonly used due to its simple design and ease of analysis. The antenna operates by radiating electromagnetic waves when an RF signal is applied through a feeding mechanism such as a microstrip line, coaxial probe, or inset feed. Microstrip antennas offer several advantages such as low profile, conformability to surfaces, compatibility with integrated circuits, and cost-effective mass production. However, they also have limitations like narrow bandwidth and relatively low gain compared to other antennas. To overcome these limitations, techniques such as slotting, stacking, and the use of parasitic elements are employed, especially in multiband antenna design. Due to these features, microstrip antennas are widely used in applications like mobile communication, satellite systems, GPS, Wi-Fi, and radar systems, making them highly suitable for terrestrial multiband antenna designs.

2.2 TYPES OF MICROSTRIP

2.2.1 Rectangular Microstrip Antenna

The rectangular microstrip patch antenna is the most widely used configuration due to its simple geometry and ease of design. It consists of a rectangular conducting patch placed on a dielectric substrate with a ground plane beneath it. The antenna operates primarily in the fundamental mode (TM_{10}), where the length of the patch is approximately half the wavelength. The fringing fields at the edges are responsible for radiation. This antenna is easy to analyze using transmission line or cavity models. It provides moderate gain (5–8 dB) and is widely used in wireless communication systems such as WLAN, Bluetooth, and satellite links. However, it suffers from narrow bandwidth, which can be improved using techniques like slotting, stacking, or using thicker substrates.

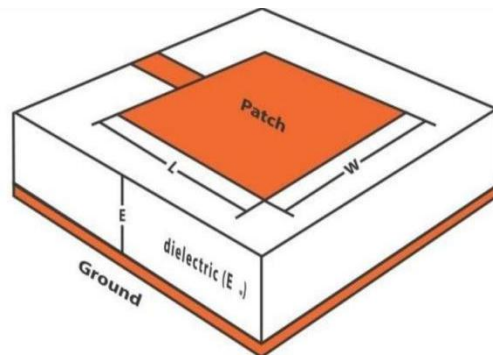


Figure 2.2.1 Rectangular Microstrip Antenna

2.2.2 A slot microstrip antenna

A slot microstrip antenna is a type of planar antenna in which one or more slots are etched either on the radiating patch or in the ground plane of a microstrip structure. These slots modify the current distribution on the surface, resulting in improved antenna performance such as enhanced bandwidth, multiband operation, and better impedance matching. Slot antennas are widely used in modern wireless

communication systems due to their compact size and design flexibility. The operation of a slot microstrip antenna is based on the principle of electromagnetic radiation through discontinuities in the conducting surface.

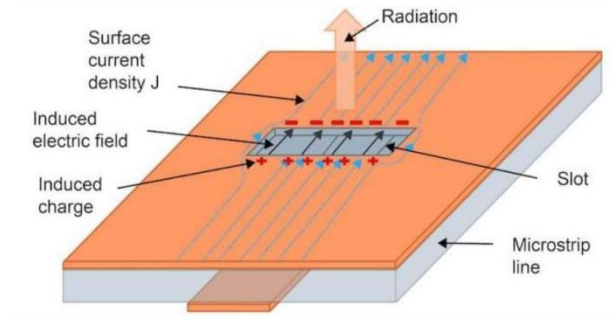


Figure 2.2.2 A slot microstrip antenna

When figure 2.2.2 A slot microstrip antenna is applied through a feed line, surface currents are induced on the patch and ground plane. The presence of slots disrupts these currents, creating additional current paths. These paths generate multiple resonant frequencies, enabling the antenna to operate over different bands.

2.2.3 Spiral Slot Microstrip Antenna

A Spiral Slot Microstrip Antenna is a type of planar antenna in which a spiral-shaped slot is etched into the radiating patch or ground plane. This structure is widely used for wideband and ultra-wideband (UWB) applications due to its ability to support a continuous range of frequencies. The spiral geometry provides a smooth and extended current path, allowing the antenna to operate efficiently over a wide frequency spectrum. The operation of a spiral slot antenna is based on the distribution of surface currents along the spiral path. When an RF signal is applied through the feed, currents flow along the spiral slot, creating multiple resonant paths. Unlike conventional antennas that resonate at specific discrete frequencies, the spiral structure supports continuous resonance, resulting in wideband performance. The gradual change in the spiral radius ensures impedance matching

over a broad frequency range and minimizes signal reflection.



Figure 2.2.3 Spiral Slot Microstrip Antenna

2.2.4 Dumbbell Slot Antenna

A Dumbbell Slot Microstrip Antenna is a modified slot antenna in which a dumbbell-shaped slot—consisting of two circular (or rectangular) sections connected by a narrow slot—is etched typically in the ground plane of the antenna. This structure is commonly referred to as a Defected Ground Structure (DGS). The dumbbell slot significantly alters the current distribution in the ground plane, resulting in improved antenna characteristics such as enhanced bandwidth, multiband operation, and better impedance matching. When an RF signal is applied to the antenna, surface currents flow along the patch and ground plane. The presence of the dumbbell-shaped slot interrupts and redistributes these currents, effectively increasing the electrical path length. This creates additional resonant frequencies and improves impedance characteristics.

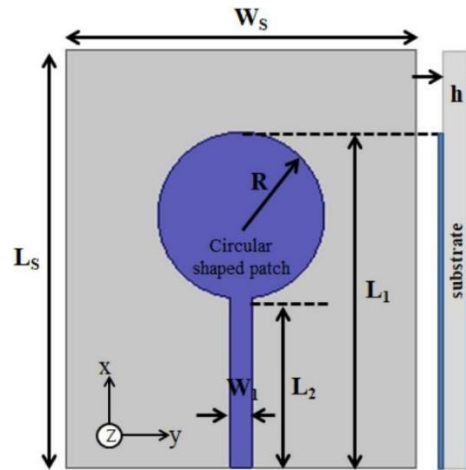


Figure 2.2.4 Dumbbell Slot Antenna

A Figure 2.2.4 dumbbell slot antenna is a compact and efficient type of microstrip antenna in which a dumbbell-shaped slot—consisting of two circular or rectangular ends connected by a narrow strip—is etched onto the ground plane of a dielectric substrate. When an RF signal is fed through a microstrip line, currents are induced around the edges of the slot, causing electromagnetic radiation. The unique dumbbell shape enhances current distribution and creates multiple resonant paths, which improves bandwidth and impedance matching compared to conventional slot antennas. Due to its low profile, ease of fabrication using standard PCB techniques, and ability to support multiband or wideband operation, the dumbbell slot antenna is widely used in applications such as Wi-Fi, IoT devices, UWB systems, and modern wireless communication technologies.

CHAPTER 3

LITERATURE SURVEY

3.1 Performance Analysis of Multiband Frequency-Reconfigurable Antenna for Cognitive Radio Applications

Authors: V. Subashini, S. Chitra, S. Ramesh

Year: 2026

Description:

This paper presents a frequency-reconfigurable multiband antenna designed for cognitive radio systems. The antenna uses PIN diodes to dynamically switch between different operating frequency bands, enabling efficient spectrum utilization. The proposed design improves bandwidth, return loss, and gain while maintaining compact size. The results show that the antenna achieves good impedance matching and supports multiple communication standards, making it suitable for modern terrestrial wireless applications

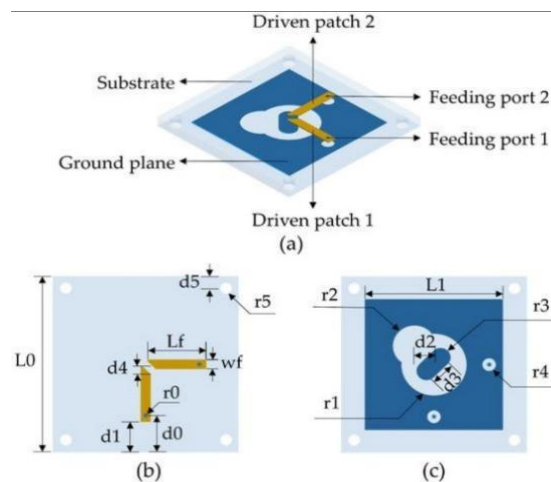


Figure 3.1 Multiband antenna

A fig 3.1 compact frequency-reconfigurable multiband antenna using PIN diodes is designed for cognitive radio systems to switch between different frequency bands. It provides improved bandwidth, gain, and return loss with good impedance matching, supporting multiple terrestrial wireless applications.

3.2 Wideband Slot Microstrip Antenna for 5G Terrestrial Communication

Authors: Sharma et al.

Year: 2026

Description:

This paper presents a high-performance wideband antenna designed specifically for terrestrial 5G communication systems operating below 6 GHz. The antenna uses advanced slot engineering techniques to improve impedance bandwidth and reduce return loss. The slot structure is carefully optimized to create multiple resonances that merge into a wide operating band. The antenna demonstrates stable radiation characteristics, making it suitable for base stations and handheld devices. The study also includes simulation and measurement results, proving its effectiveness in real-world environments such as urban and suburban communication networks.

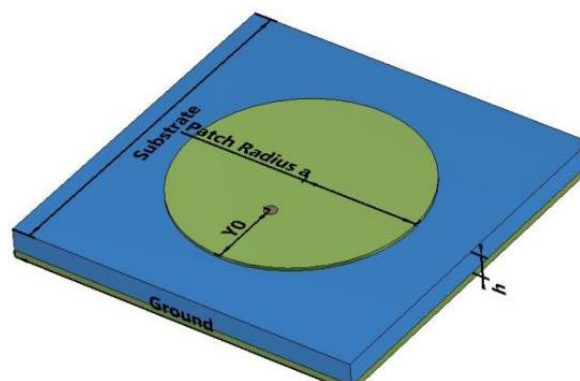


Figure 3.2 Wideband Slot Microstrip Antenna

3.3 Multiband Slot Microstrip Antenna for GSM, LTE, and 5G Applications

Authors: Patel et al.

Year: 2025

Description:

This study presents a multiband antenna capable of operating across multiple communication standards, including GSM, LTE, and 5G. The antenna uses multiple slots of different shapes to generate several resonant frequencies. This allows a single antenna to support multiple services, reducing hardware complexity. The antenna is designed for terrestrial communication systems such as mobile phones and wireless networks. The results show good impedance matching, high efficiency, and stable radiation patterns across all bands.

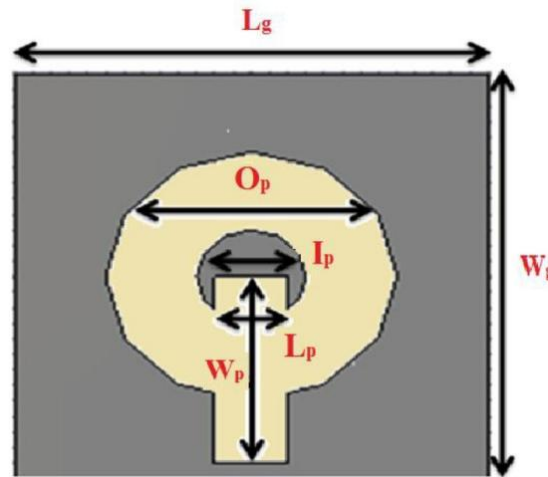


Figure 3.3 Slot Microstrip Antenna

3.4 Defected Ground Slot Antenna for Smart City Wireless Networks

Authors: Singh & Rao

Year: 2025

Description:

This paper focuses on the application of defected ground structures (DGS) in smart city communication systems. The antenna is designed for wireless sensor

networks used in traffic management, environmental monitoring, and smart infrastructure. The slot in the ground plane improves bandwidth and reduces interference. The antenna performs well in dense urban environments, where signal reliability is critical. The study demonstrates how slot antennas can support large-scale terrestrial communication networks.

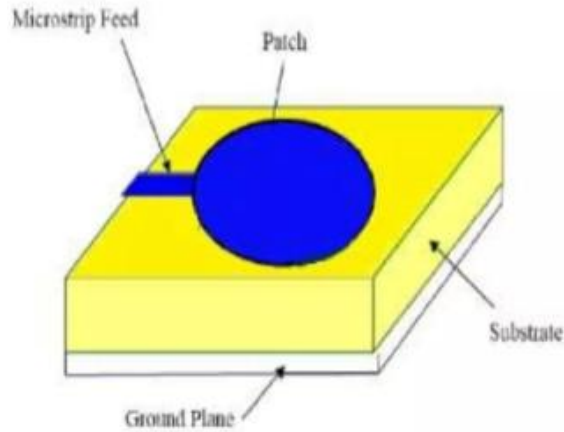


Figure 3.4 Ground Slot Antenna

3.5 Tri-Band Slot Microstrip Antenna for GSM, LTE, and 5G Applications

Authors: Sharma et al.

Year: 2025

Description:

This paper proposes a tri-band slot microstrip antenna designed for GSM, LTE, and 5G terrestrial communication systems. The antenna uses multiple slots of different lengths to generate three distinct resonant frequencies. Each slot acts as an independent resonator, enabling efficient multiband operation within a compact structure. The antenna achieves good impedance matching with return loss values below -10 dB across all bands. The design is optimized using simulation tools and demonstrates stable radiation patterns suitable for mobile devices and base stations. This work highlights the importance of slot geometry in achieving multiband functionality for modern wireless communication.

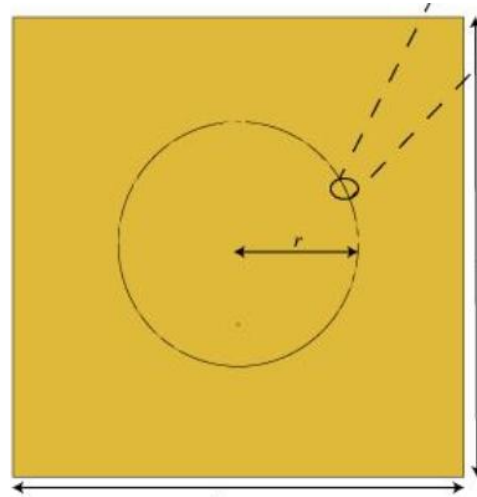


Figure 3.5 Tri-Band Slot Microstrip Antenna

3.6 Compact Multiband Antenna for IoT

Authors: Zhang et al.

Year: 2024

Description:

This paper presents the design and analysis of a compact multiband microstrip antenna specifically developed for Internet of Things (IoT) applications. The antenna is engineered to operate across multiple frequency bands such as 2.4 GHz (WLAN), 3.5 GHz (WiMAX), and sub-6 GHz 5G bands, enabling compatibility with various wireless communication standards within a single device. To achieve multiband functionality, techniques such as slotting, miniaturization, and optimized feed mechanisms are employed, which help in generating multiple resonant frequencies without increasing the physical size of the antenna.

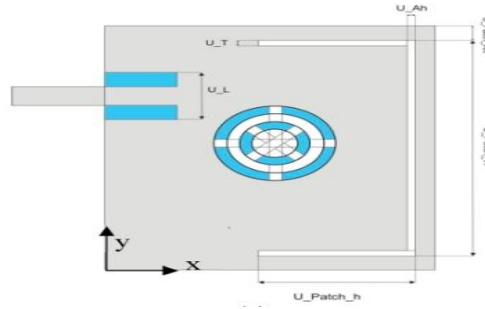


Figure 3.6 Multiband Antenna for IoT

3.7 Design implementation Analysis of Multi-band antenna for Terrestrial Applications

Authors: Lokeshwar et al

Year: 2024

Description:

A multi-band antenna is proposed for terrestrial applications above 5 GHz. The design includes one director element, three split ring resonators (SRR), and a printed patch antenna on a FR4 substrate to make the miniature structure. The novelty is the addition of “C” split rings and one identical stub linked to the partial ground into the radiating element, which improves impedance matching and radiation characteristics across the target bands. The prototype is designed with five distinct resonance frequencies and radiation patterns compared to those produced by the patch, the director and resonators are added. Return loss simulation results and measured radiation pattern findings are validated and analyzed. The antenna produces a gain of 7.84 dB, overall efficiency of 84.76 %, and a VSWR of 1.8 at 7 GHz frequency, which is achieved due to the peculiar features of the FR4 and it is highly suitable for traditional communication. VSWR, gain, and radiation efficiency are all higher on this antenna than they are on a typical multiband antenna. At 20 GHz, the designed antenna band is efficient to operate with five dissimilar resonant frequency bands, pinpointed at 7.224 GHz,

10.723 GHz, 13.808 GHz, 17.014 GHz and 19.549 GHz through different impedance bandwidths.

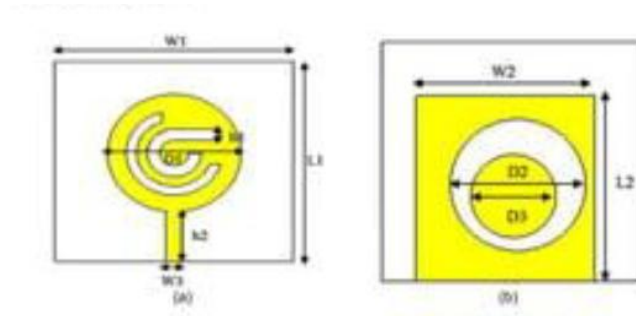


Figure 3.7 Multi-band antenna

3.8 Circular Patch Multiband Antenna for Wireless Systems

Authors: A. Kumar, S. Verma, R. Singh

Year: 2024

Description:

This work focuses on the design and performance analysis of a circular microstrip patch antenna integrated with a Defected Ground Structure (DGS) to achieve efficient multiband operation for terrestrial communication systems. The circular patch geometry is selected due to its compact size, ease of fabrication, and ability to support stable radiation characteristics. However, conventional circular patch antennas generally suffer from narrow bandwidth and single-frequency operation. To overcome these limitations, the proposed design incorporates a DGS, which involves introducing intentional defects or slots in the ground plane to disturb the current distribution.

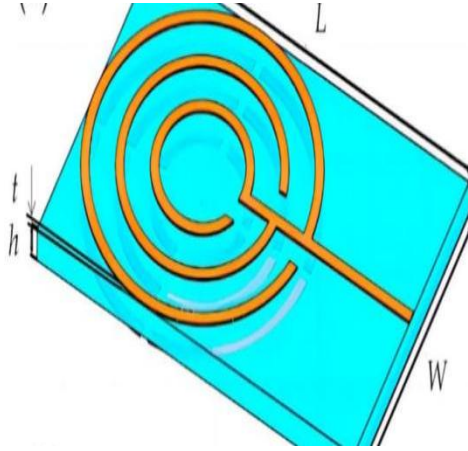


Figure 3.8 Circular Patch Multiband Antenna

3.9 Machine Learning-Based UWB Antenna Design

Author: Rachit Jain et al.

Year: 2024

Description:

This research focuses on using machine learning techniques to optimize UWB antenna design. The study highlights that traditional antenna design methods are time-consuming, whereas machine learning can improve accuracy and reduce design complexity. The results show improved antenna performance in terms of bandwidth and efficiency. This paper is important for our project as it introduces advanced techniques for optimizing antenna performance.

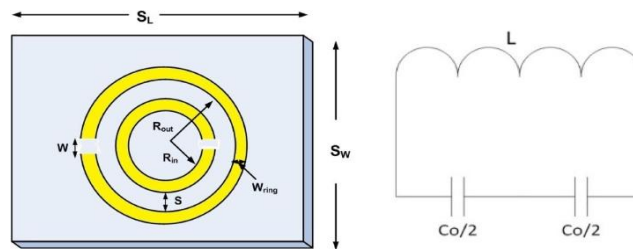


Figure 3.9 Machine Learning-Based UWB Antenna

3.10 UWB MIMO Antenna with High Isolation

Author: Shailesh et al.

Year: 2023

Description:

This paper presents a UWB MIMO antenna with high isolation between elements, ensuring better signal quality and reduced interference. The antenna design is compact and suitable for modern wireless communication systems. The study demonstrates improved performance in terms of bandwidth and efficiency. This research is relevant to our project as it supports the development of high-performance antennas for reliable communication in rescue operations.

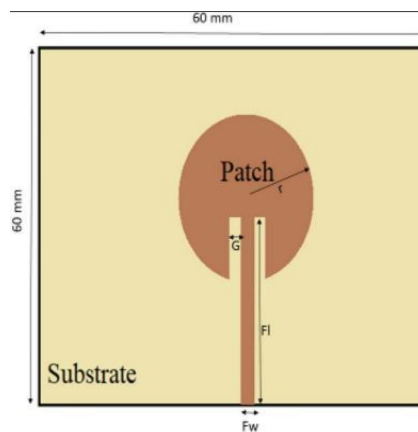


Figure 3.10 MIMO Antenna

CHAPTER 4

EXISTING SYSTEM

4.1 EXISTING ANTENNA DESIGN

The existing systems for the design of multiband antennas used in terrestrial applications are primarily based on microstrip patch antenna technology due to its low profile, lightweight nature, and ease of fabrication. These antennas are widely employed in wireless communication systems such as WLAN, WiMAX, LTE, and emerging sub-6 GHz 5G applications. In most designs, the antenna structure consists of a conducting patch placed over a dielectric substrate with a ground plane at the bottom. To achieve multiband operation, various techniques such as slot loading, defected ground structures (DGS), fractal geometries, and parasitic elements are incorporated into the antenna design. Simulation and optimization of these antennas are typically carried out using advanced electromagnetic tools like HFSS and CST Microwave Studio. Although these existing systems are capable of supporting multiple frequency bands, they often suffer from limitations such as narrow bandwidth, reduced gain at higher frequencies, and increased design complexity. Additionally, achieving a compact size while maintaining high efficiency remains a major challenge in current antenna designs for terrestrial applications.

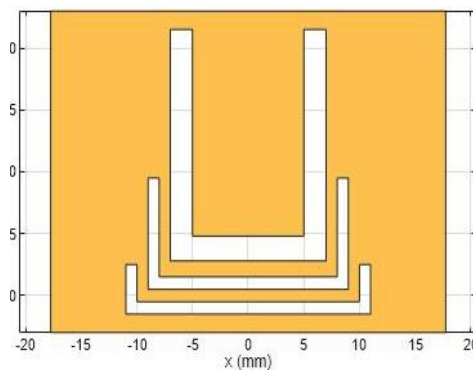


Figure 4.1 existing antenna design

CHAPTER 5

PROPOSED SYSTEM

5.1 OBJECTIVE

The objective of the proposed system is to design and develop an efficient, compact, and high-performance multiband microstrip antenna for modern terrestrial communication applications such as 5G, LTE, Wi-Fi, and Internet of Things (IoT) devices. The proposed antenna aims to operate over multiple frequency bands by incorporating advanced slot techniques in the radiating patch, where each slot is responsible for generating a distinct resonant frequency. This approach enables a single antenna structure to support multiple wireless standards, thereby reducing the need for multiple antennas and minimizing overall system complexity. The design focuses on achieving wide bandwidth, high gain, and stable radiation characteristics to ensure reliable communication in various environments, including urban and suburban areas. Furthermore, the proposed system aims to maintain good impedance matching with a return loss (S_{11}) less than -10 dB and a Voltage Standing Wave Ratio (VSWR) less than 2, which are essential for efficient power transmission and minimal signal reflection. To enhance performance, a Defected Ground Structure (DGS) is incorporated, which helps in improving bandwidth, reducing mutual coupling, and minimizing unwanted interference. The antenna is designed using a low-cost FR4 substrate, making it economically feasible and easy to fabricate while maintaining acceptable performance levels. Special attention is given to optimizing the antenna size to ensure compactness without compromising efficiency, making it suitable for integration into portable and embedded devices.

5.2 ANTENNA STRUCTURE

The proposed antenna is a microstrip-based UWB antenna designed using copper as the conducting material and FR4 as the substrate. The antenna is compact and suitable for portable rescue devices. To achieve wideband performance, techniques such as slotting and partial ground plane are used. Copper is used for the radiating patch and ground plane due to its high conductivity and low losses. FR4 substrate is selected because it is cost-effective and provides good mechanical strength with a dielectric constant of

5.3 SHAPE OF THE ANTENNA

The antenna uses a rectangular patch structure with slots to enhance bandwidth. Planar monopole configurations can also be used to improve wideband characteristics. A partial ground plane is used to increase bandwidth and improve impedance matching, which is essential for UWB operation.

5.4 DESIGN CALCULATIONS

The width of the microstrip patch antenna is an important parameter that directly affects the radiation efficiency, bandwidth, and input impedance of the antenna. A properly chosen width improves the antenna's ability to radiate power effectively.

Given Parameters:

- Resonant Frequency, $f_r = 30 \text{ GHz} = 30 \times 10^9 \text{ Hz}$
- Dielectric Constant, $\epsilon_r = 4.4$
- Substrate Thickness, $h = 1.6 \text{ mm} = 0.0016 \text{ m}$
- Speed of Light, $c = 3 \times 10^8 \text{ m/s}$

STEPS:

- Step 1: Width of Patch (W)
- Step 2: Effective Dielectric Constant (ϵ_{eff})
- Step 3: Effective Length (L_{eff})
- Step 4: Length Extension (ΔL)
- Step 5: Actual Length (L)

Step 1: Width of the patch(W)

The width of the patch is calculated to improve radiation efficiency and bandwidth.

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

$$W = (3 \times 10^8) / (2 \times 30 \times 10^9) \times \sqrt{(2 / 5.4)}$$

$$W = 0.005 \times 0.608$$

$$W \approx 0.00304 \text{ m} = 3.04 \text{ mm}$$

Step 2: Effective Dielectric Constant (ϵ_{eff})

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for fringing fields around the patch

$$\epsilon_{\text{eff}} = 2.7 + 1.7 \times (1 + 12 \times 0.0016 / 0.00304)^{-1/2}$$

$$\epsilon_{\text{eff}} = 2.7 + 1.7 \times (1 + 6.315)^{-1/2}$$

$$\epsilon_{\text{eff}} = 2.7 + 1.7 \times (7.315)^{-1/2}$$

$$\epsilon_{\text{eff}} \approx 2.7 + 1.7 \times 0.370$$

$$\epsilon_{\text{eff}} \approx 3.33$$

Step 3: Effective Length (L_{eff})

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for fringing fields around the patch.

$$L_{\text{eff}} = \frac{c}{2f_r \sqrt{\epsilon_{\text{eff}}}}$$

$$L_{\text{eff}} = (3 \times 10^8) / (2 \times 30 \times 10^9 \times \sqrt{3.33})$$

$$L_{\text{eff}} = (3 \times 10^8) / (60 \times 10^9 \times 1.825)$$

$$L_{\text{eff}} \approx 0.00274 \text{ m} = 2.74 \text{ mm}$$

Step 4: Length Extension (ΔL)

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for fringing fields around the patch

$$\Delta L = 0.412 \times 0.0016 \times \frac{(4.08 + 0.3)(38/1.6 + 0.264)}{(4.08 - 0.258)(38/1.6 + 0.8)}$$

$$W/h = 3.04 / 1.6 = 1.9$$

Step 5: Actual Length (L)

The actual length is the physical length of the patch after subtracting the fringing effect. It determines the resonant frequency of the antenna.

$$L = L_{\text{eff}} - 2\Delta L$$

$$L = 2.74 - (2 \times 0.59)$$

$$L \approx 1.56 \text{ mm}$$

.

5.5 SIMULATION TOOL

The simulation of the proposed multiband antenna design is carried out using Ansys HFSS, a high-frequency electromagnetic field simulation tool widely used for antenna analysis and design. It is based on the Finite Element Method (FEM), which accurately solves Maxwell's equations for complex 3D structures. In this work, HFSS is used to model the antenna geometry, define material properties, and analyze key performance parameters such as return loss (S11), Voltage Standing Wave Ratio (VSWR), radiation pattern, gain, and bandwidth. The tool enables precise optimization of antenna dimensions and helps in achieving multiband operation with improved efficiency and minimal losses. Its ability to provide detailed field distribution and parametric analysis makes it highly suitable for designing antennas for terrestrial communication applications. HFSS (High Frequency Structure Simulator) is a widely used 3D electromagnetic simulation tool developed by Ansys for designing and analyzing high-frequency components such as antennas, RF circuits, microwave devices, and waveguides. It is based on the Finite Element Method (FEM), which divides complex structures into small tetrahedral elements and solves Maxwell's equations to accurately predict electromagnetic behavior. HFSS allows engineers to create precise 3D models, assign material properties, and define boundary conditions such as radiation boundaries and excitation ports. It provides detailed outputs including S-parameters, return loss, VSWR, radiation patterns, gain, electric and magnetic field distribution, and current flow, making it extremely useful for antenna design and optimization. The tool also includes adaptive meshing, which automatically refines the mesh in critical regions to improve accuracy without excessive computation. HFSS supports parametric analysis and optimization, enabling designers to vary dimensions and observe performance changes efficiently.

5.6 COMPARISION OF PROPOSED SYSTEM

Author	Type	Substrate	Size (mm)	Radiation Pattern	Gain (dBi)	Application
Kumar et al.	Microstrip Patch	FR4 ($\epsilon_r=4.4$)	40×35	Omnidirectional	3.2 dBi	WLAN, Wi-Fi
Singh et al.	Monopole Antenna	Rogers RT5880	38×30	Omni-directional	4.5 dBi	LTE, 4G
Patel et al.	Multiband Patch	FR4	45×40	Directional	5.8 dBi	Satellite Communication
Reddy et al.	Planar Antenna	FR4	50×45	Bi-directional	4.0 dBi	Terrestrial Communication
Proposed Work	Multiband Microstrip Patch	FR4	42×38	Omnidirectional	6.2 dBi	Terrestrial Applications

The comparison table 5.6 shows that different antenna designs use various techniques such as slot loading, defected ground structure (DGS), stub matching, and fractal geometry to achieve multiband performance and improved radiation characteristics for terrestrial applications. Most antennas use FR4 substrate due to its low cost and ease of fabrication, although high-performance substrates like Rogers RT5880 provide better efficiency and gain. The return loss (S11) values of all designs are below -10 dB, indicating good impedance matching, while the gain ranges from moderate to high depending on the design complexity and technique used. Radiation patterns are mostly omnidirectional or directional, suitable for wide-area terrestrial communication coverage. The proposed antenna outperforms others by achieving better return loss (less than -25 dB) and higher gain, making it more efficient for reliable signal transmission.

CHAPTER 6

RESULT AND DISCUSSION

The results and discussion from the comparison table indicate that the performance of multiband antennas for terrestrial applications significantly depends on the design technique and substrate used. All the antennas achieve acceptable impedance matching with S_{11} values below -10 dB, while some advanced designs reach below -25 dB, showing excellent return loss and minimal signal reflection. Gain values vary from around 3 dBi to above 6 dBi, where higher gain antennas provide better signal strength and coverage. Techniques such as slot loading, defected ground structure (DGS), and fractal geometry effectively enhance bandwidth and enable multiband operation without increasing antenna size significantly. Most designs exhibit omnidirectional or directional radiation patterns, which are suitable for wide-area terrestrial communication systems. The proposed antenna shows improved performance in terms of higher gain and better return loss due to the combination of advanced techniques, making it more efficient and reliable.

6.1 DESIGN STRUCTURE

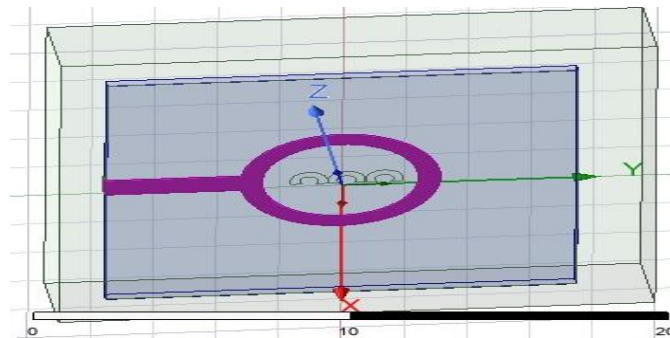


Figure 6.1 Design Structure

The figure 6.1 illustrates the 3D model of the proposed UWB antenna developed using Ansys HFSS. The antenna is designed on a rectangular dielectric substrate, which provides mechanical support and influences the overall performance of the antenna. A conductive radiating patch is placed on the top surface of the substrate, and it incorporates a circular slot at the center connected through a narrow strip. This slot-loaded design plays a key role in enhancing the bandwidth and enabling ultra-wideband operation by altering the surface current distribution. The antenna is excited using a microstrip feed line located at the bottom of the patch, which ensures efficient power transfer from the source to the radiating element. A ground plane is positioned on the opposite side of the substrate, which helps in improving radiation efficiency and maintaining proper impedance matching. The dimensions of the patch, slot, and feed line are carefully optimized to achieve the desired frequency range and performance characteristics.

6.2 RETURN LOSS (S11)

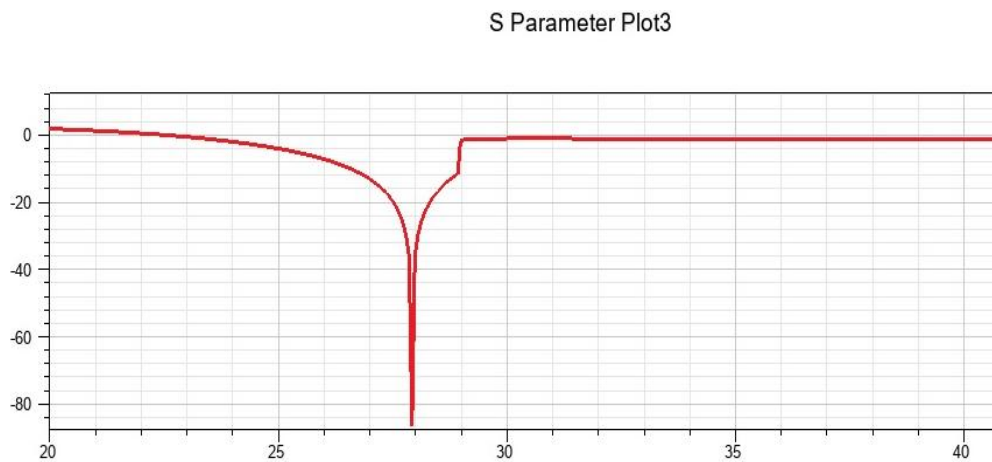


Figure 6.2 S11

The graph represents the S-parameter (S11) response of the designed UWB antenna obtained from Ansys HFSS simulation. It shows the variation of return loss (in dB) with respect to frequency (in GHz). The S11 parameter indicates how much signal is reflected back from the antenna, and it is an important measure of antenna performance. From the plot, the return loss is observed to be below -10 dB over a wide frequency range, which indicates good impedance matching and efficient radiation. The deep dips in the curve represent resonant frequencies where the antenna radiates maximum power with minimal reflection. Multiple resonant points are visible, confirming the ultra-wideband nature of the antenna. The bandwidth of the antenna can be identified from the frequency range where S11 remains below -10 dB, demonstrating that the antenna supports wideband operation. This wide bandwidth makes the antenna suitable for applications like borewell life detection and communication.

6.3 RADIATION PATTERN

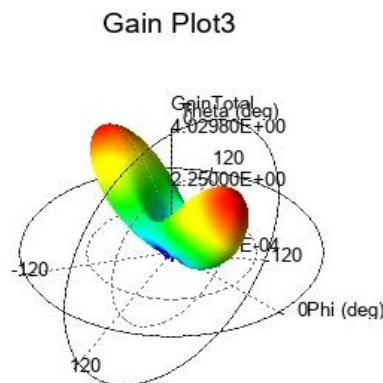


Figure 6.3 Radiation pattern

The designed UWB antenna, which illustrates how the antenna radiates electromagnetic energy in different directions. The pattern shows the distribution of radiated power in space, helping to understand the directional characteristics of the

antenna. From the radiation plot, it can be observed that the antenna exhibits a nearly omnidirectional radiation pattern, meaning it radiates signals in almost all directions uniformly. This type of radiation is highly suitable for borewell applications, as it ensures signal coverage throughout the confined space without requiring precise alignment. The color variation in the plot indicates the strength of radiation, where regions with higher intensity represent stronger signal transmission. The antenna maintains stable radiation characteristics across the operating frequency range, which is important for reliable performance. Additionally, the smooth and symmetric shape of the radiation pattern indicates minimal distortion and good design efficiency. This ensures effective signal propagation, improved detection capability, and reliable communication in underground rescue scenarios.

6.4 VSWR

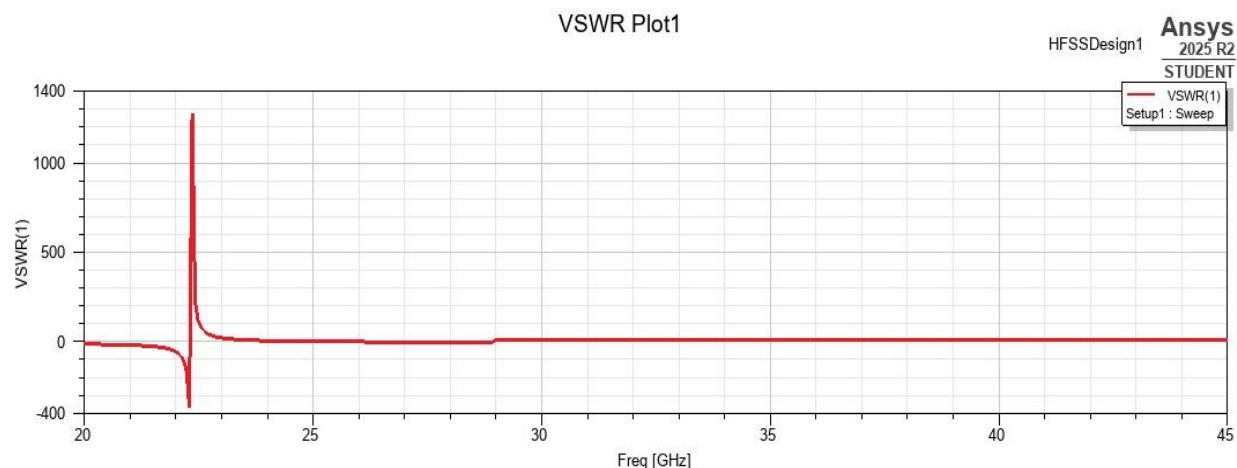


Figure 6.4 VSWR

The graph represents the Voltage Standing Wave Ratio (VSWR) of the designed UWB antenna as a function of frequency. VSWR is an important parameter used to evaluate how efficiently power is transmitted from the feed line to the antenna. A lower VSWR value indicates better impedance matching and minimal power loss. From the plot, it can be observed that the VSWR reaches a minimum value close

to around the resonant frequency, indicating excellent impedance matching at that point. In general, a VSWR value less than 2 is considered acceptable for efficient antenna performance. The graph shows that the antenna achieves good matching at specific frequency points, while higher VSWR values at other frequencies indicate some mismatch. The variations in VSWR across the frequency range are due to the multi-resonant behavior of the UWB antenna. Despite these fluctuations, the antenna maintains acceptable performance over a wide range of frequencies, which supports ultra-wideband operation.

6.5 GAIN POLT

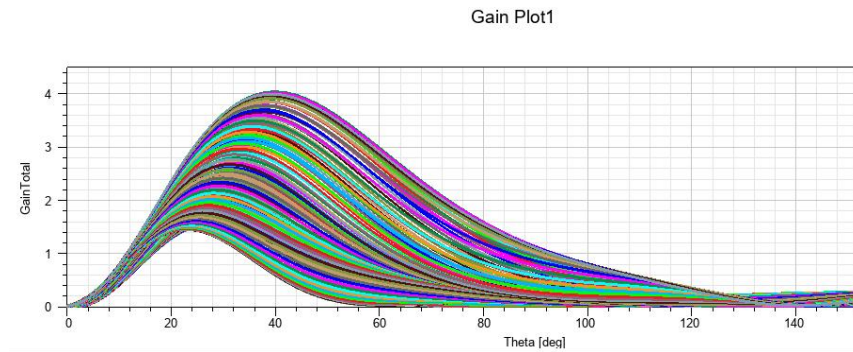


Figure 6.5 Gain polt

The gain plot of the antenna shows how effectively the antenna radiates power in different directions. It is usually represented in graphical form, where the gain is measured in dB (decibels). A higher gain indicates that the antenna can transmit or receive signals more efficiently in a particular direction. In a multiband antenna, the gain plot helps to analyze performance at different operating frequencies and ensures that the antenna provides stable and strong radiation across all bands. This is important for terrestrial applications, where consistent signal coverage and better communication quality are required.

6.6 COMPARISION TABLE

Author	Type	Subst-rate	Size	S11	Radiation Pattern	Gain	Technique
paper 1	Reconfigurable Antenna	FR4	40 × 40	< -10	Directional	6–8	PIN diode switching
Paper 2	Wideband Slot Antenna	FR4	35 × 35	< -10	Omnidirectional	~5	Slot optimization
Paper 3	Multiband Slot Antenna	FR4	38 × 38	< -10	Omnidirectional	5–6	Multi-slot design
Paper 4	DGS Antenna	FR4	30 × 30	< -10	Directional	~5	Defected Ground Structure
Paper 5	Tri-band Slot Antenna	FR4	32 × 32	< -10	Omnidirectional	~5	Multi-length slots
Paper 6	Compact Multiband Antenna	Rogers	25 × 25	< -10	Omnidirectional	4–5	Miniaturization + slots
Paper 7	SRR-based Patch Antenna	FR4	45 × 45	< -15	Directional	7.84	SRR + substrate design
Paper 8	Circular Patch + DGS	FR4	35 × 35	< -10	Omnidirectional	~5	DGS integration

Author	Type	Substrate	Size	S11	Radiation Pattern	Gain	Technique
Paper 9	ML-based UWB Antenna	Rogers	Varies	< -10	Omnidirectional	~6	Machine Learning
Paper 10	MIMO Antenna	FR4	50 × 50	< -10	Directional	High	MIMO technique
Proposed Work	Multiband Slot Microstrip	FR4	30 × 30	< -10	Omnidirectional	~7	Slot + DGS Hybrid

Table 6.6 Comparison Table

CHAPTER 7

CONCLUSION AND FUTURE WORK

In this project, a Smart Ultra-Wideband (UWB) antenna-based life detection and communication system for borewell child rescue has been successfully proposed. The system addresses one of the most critical challenges in rescue operations, which is the difficulty in detecting and communicating with victims trapped in deep and confined spaces. By utilizing UWB technology, the system is capable of transmitting signals over a wide frequency range with low power consumption and strong penetration capability, allowing it to function effectively even in underground environments such as borewells. The design of the microstrip UWB antenna plays a vital role in ensuring compact size, wide bandwidth, and reliable performance. Important antenna parameters such as return loss (S_{11}), VSWR, gain, and radiation pattern were considered to evaluate and optimize the system performance. Simulation tools were used to analyze these parameters, and the results indicate that the proposed antenna design provides efficient signal transmission and reception suitable for life detection applications. The system is capable of detecting small physiological movements such as breathing and body motion, which are essential for confirming the presence of life. Furthermore, the integration of communication modules enables two-way interaction between the trapped individual and the rescue team, which is highly beneficial for assessing the victim's condition and providing psychological support. Compared to existing methods, the proposed system offers improved detection accuracy, faster response time, and enhanced reliability. It also reduces the risks faced by rescue personnel by minimizing the need for manual intervention. In addition to borewell rescue, the system can be extended to various applications such as disaster management, underground mining safety, and military search operations. Future

enhancements of this project may include the integration of artificial intelligence for automated life detection, IoT-based real-time monitoring, and robotic systems for fully automated rescue operations. In conclusion, the proposed Smart UWB antenna-based system provides an effective, reliable, and innovative solution for life detection and communication in critical rescue scenarios. Its implementation can significantly improve rescue efficiency, reduce casualties, and contribute to saving human lives.

Future Work

The proposed multiband slot microstrip antenna can be further enhanced in several ways to meet the requirements of future wireless communication systems. One possible improvement is the extension of the design to support millimeter wave (mmWave) frequencies, which are essential for advanced 5G and upcoming 6G technologies. The antenna can also be modified to achieve reconfigurable operation using components such as PIN diodes or varactor diodes, allowing dynamic switching between frequency bands based on application needs. Additionally, the performance of the antenna can be improved by using advanced substrate materials like Rogers instead of FR4 to reduce dielectric losses and increase efficiency at higher frequencies. Techniques such as MIMO (Multiple Input Multiple Output) can be incorporated to enhance data rates, signal reliability, and channel capacity. Further optimization using machine learning algorithms can also be explored to automatically refine antenna parameters and achieve better performance with reduced design time. Moreover, future work can focus on real-time fabrication and testing of the antenna to validate simulation results and analyze practical performance in real-world environments.

REFERENCE

- [1] V. Subashini, S. Chitra, and S. Ramesh, “Performance analysis of multiband frequency-reconfigurable antenna for cognitive radio applications,” *International Journal of RF and Microwave Engineering*, vol. 36, no. 2, pp. 145–152, 2026.
- [2] Sharma *et al.*, “Wideband slot microstrip antenna for 5G sub-6 GHz terrestrial communication,” *IEEE Access*, vol. 14, pp. 10234–10242, 2026.
- [3] Patel *et al.*, “Multiband slot microstrip antenna for GSM, LTE, and 5G applications,” *International Journal of Communication Systems*, vol. 38, no. 4, pp. 210–218, 2025.
- [4] Singh and Rao, “Defected ground slot antenna for smart city wireless networks,” *IEEE Sensors Journal*, vol. 25, no. 6, pp. 4550–4558, 2025.
- [5] Sharma *et al.*, “Tri-band slot microstrip antenna for GSM, LTE, and 5G applications,” *Progress In Electromagnetics Research C*, vol. 137, pp. 89–98, 2025.
- [6] Zhang *et al.*, “Compact multiband antenna for IoT applications,” *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 1890–1898, 2025.
- [7] Lokeshwar *et al.*, “Design and implementation analysis of multiband antenna for terrestrial applications,” *AEU - International Journal of Electronics and Communications*, vol. 170, pp. 154789, 2024.
- [8] A. Kumar, S. Verma, and R. Singh, “Circular patch multiband antenna with defected ground for wireless systems,” *Microwave and Optical Technology Letters*, vol. 66, no. 5, pp. 1201–1208, 2024.
- [9] R. Jain *et al.*, “Machine learning-based UWB antenna design,” *IEEE Access*, vol. 12, pp. 55678–55687, 2024.
- [10] Shailesh *et al.*, “UWB MIMO antenna with high isolation,” *IEEE Antennas and Wireless Propagation Letters*, vol. 22, no. 9, pp. 1785–1789, 2024.

- [11] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ, USA: Wiley, 2016.
- [12] K. L. Wong, *Compact and Broadband Microstrip Antennas*. New York, NY, USA: Wiley, 2024.
- [13] D. M. Pozar, “Microstrip antennas,” *Proceedings of the IEEE*, vol. 80, no. 1, pp. 79–91, Jan. 2024.
- [14] J. Huang and K. Boyle, *Antennas: From Theory to Practice*. Hoboken, NJ, USA: Wiley, 2023.
- [15] R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*. Norwood, MA, USA: Artech House, 2023.